The Viewership Effects of Crossover Cobranding

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Shared universes and crossover events

Marvel Cinematic Universe



- Shared universe ft. Avengers, Spider-Man, and other franchises
- Large-scale crossover events
- Massive audience engagement
- Long-term franchise retention

CW Arrowverse



- Shared universe ft. Arrow, The Flash, Supergirl, and other characters
- Annual crossover events
- Sequential multi-show storytelling

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Crossovers as cobranding

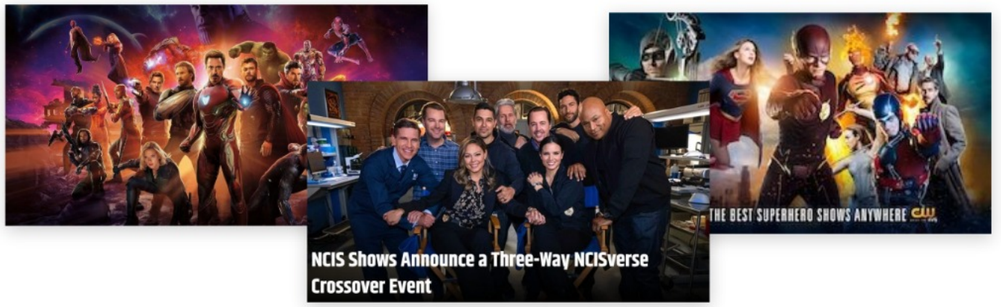
Crossovers are not only an entertainment phenomenon; they are a distinct form of cobranding.

Traditional cobranding



- Typically, 2-brand collaborations
- Focus on static product-level outcomes (e.g., sales, brand adoption)
- Effects examined during and after collaboration

Crossover cobranding



- Multi-brand/show collaborations
- Sequential & serialized audience engagement
- Responses evolve dynamically over time
- Outcomes unfold before, during, and after the event

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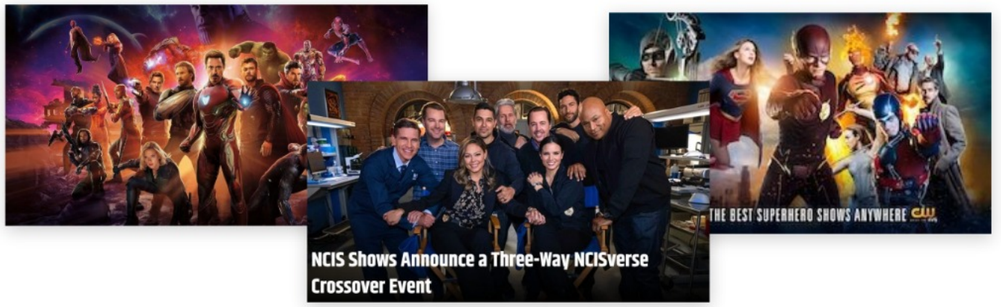
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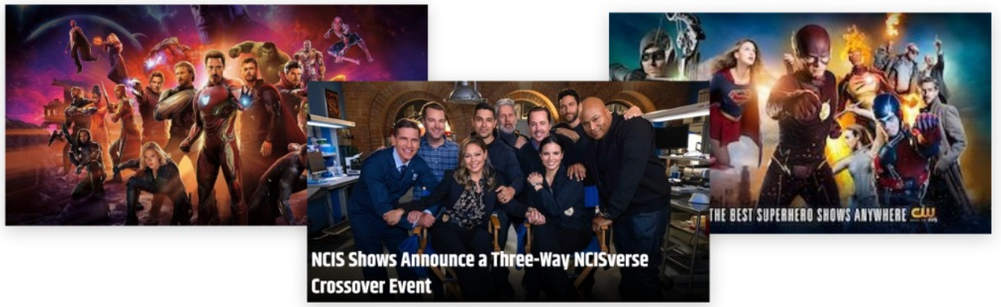
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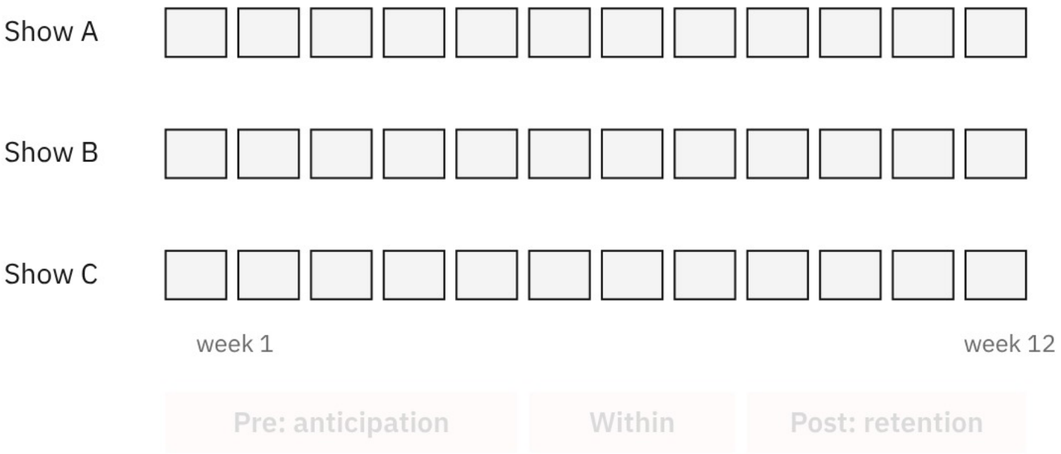
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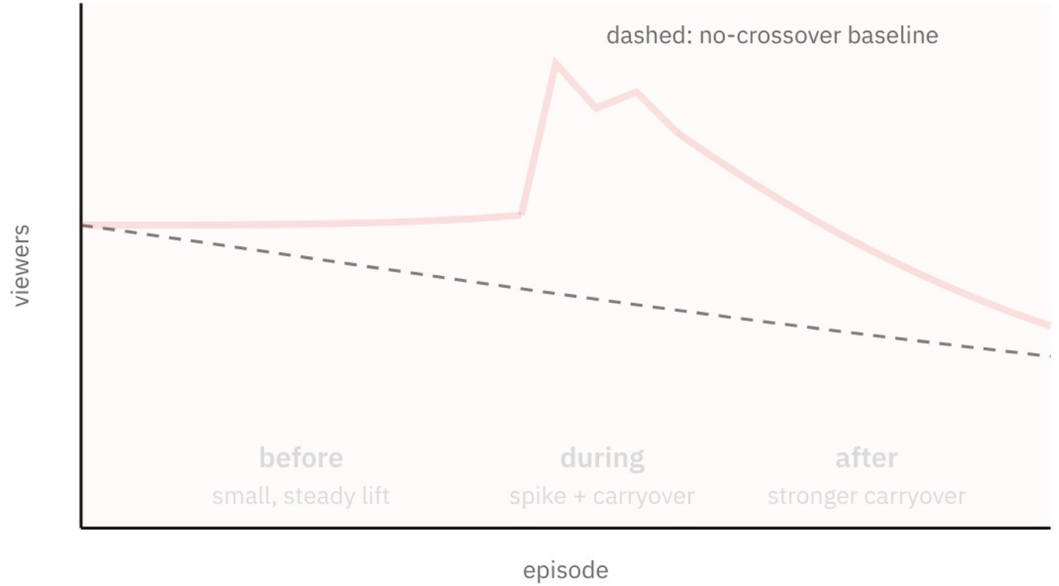
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Episode schedule (one season)



One story told across consecutive episodes of several shows, announced before the season and promoted for weeks in advance.

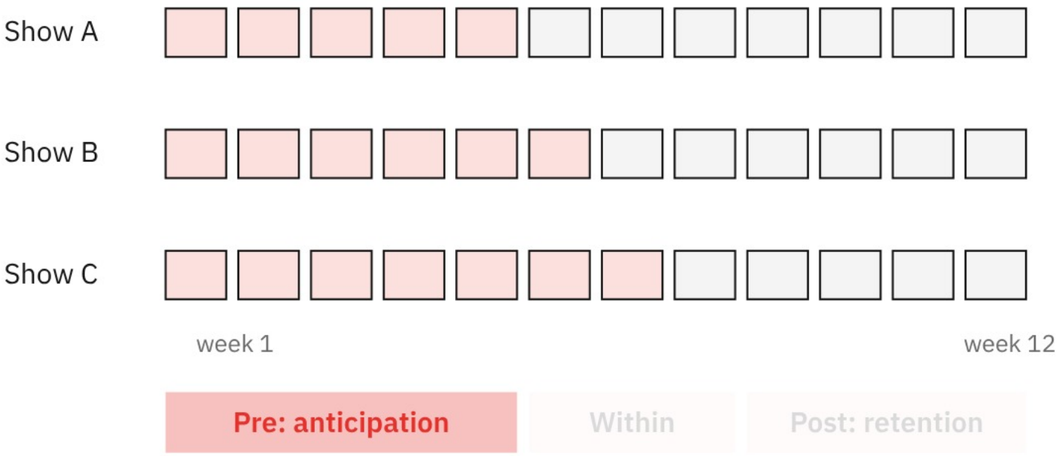
Viewership of one participating show



Illustrative. The model estimates each of the three segments separately.

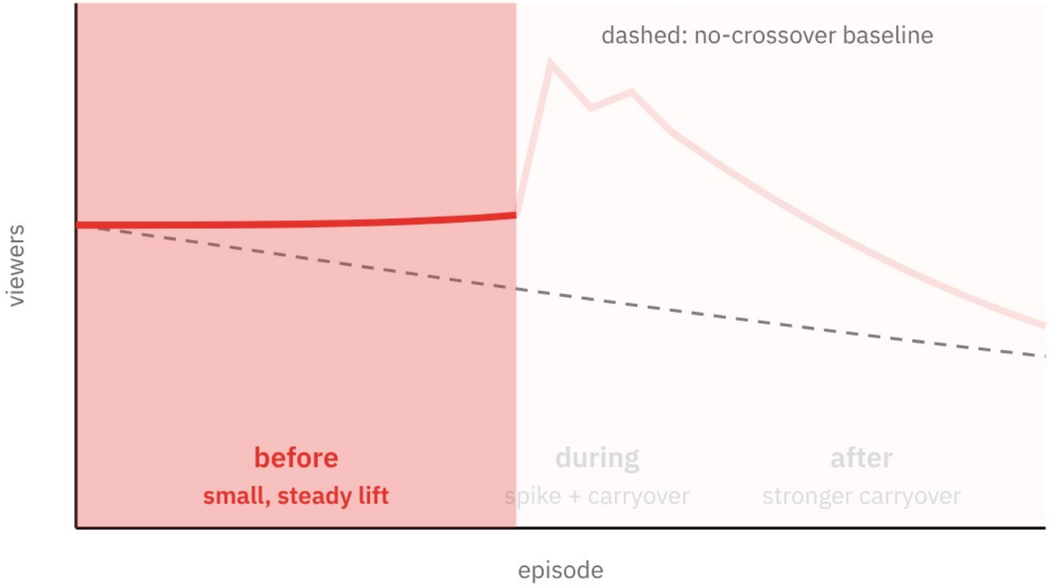
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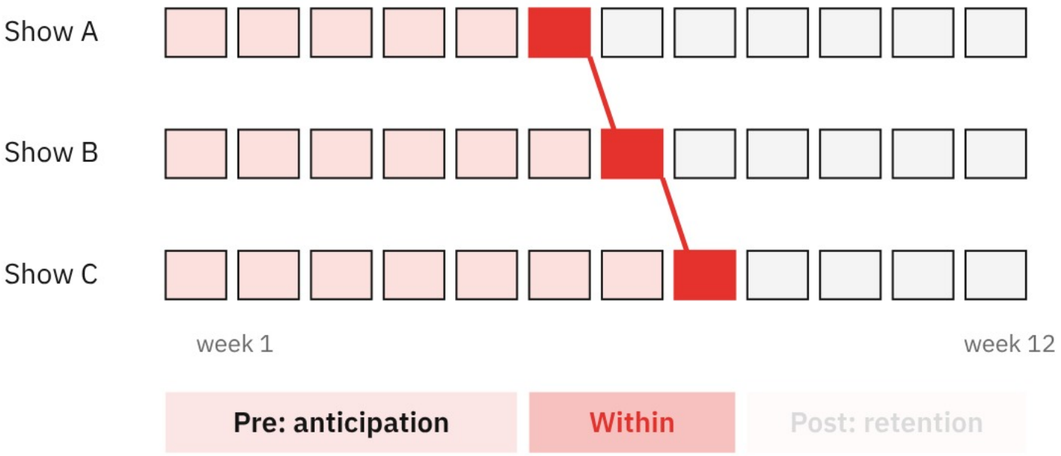
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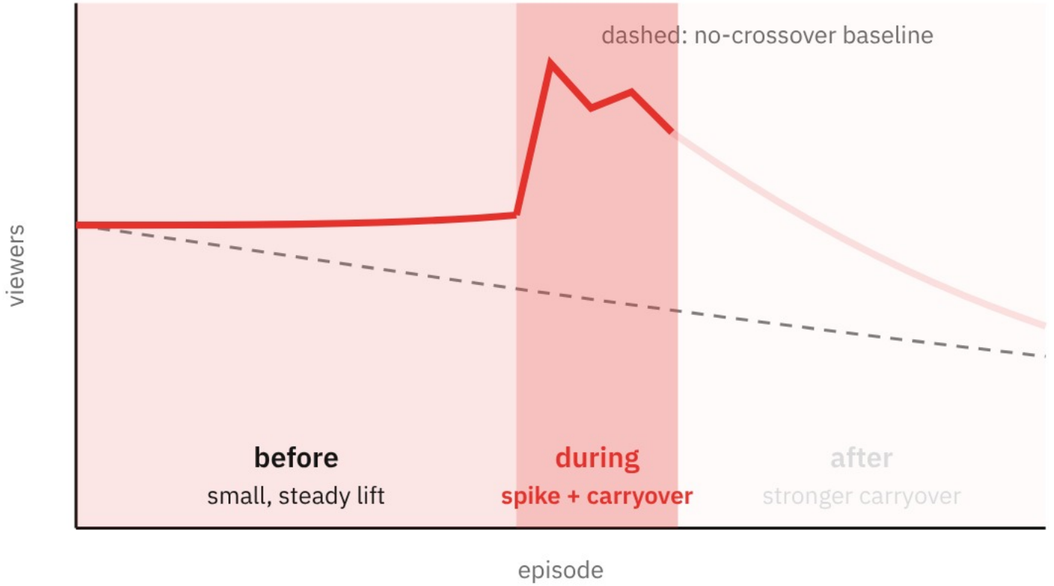
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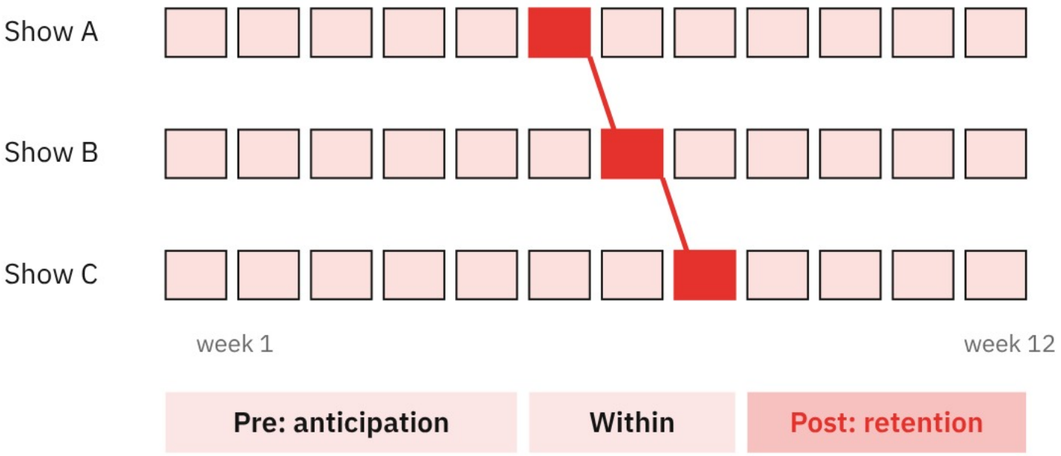
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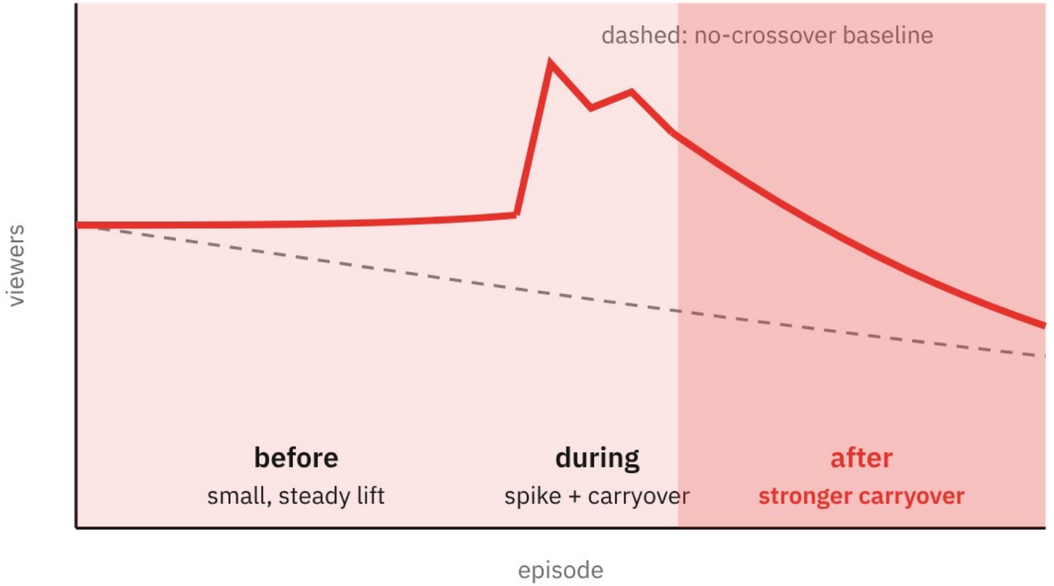
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Research questions

Before

Do crossover events increase viewership before the event through anticipation?

During

How large is the immediate spike in the first crossover episode, and how much carries across episodes?

After

Do crossover events create post-event retention?

Moderator: relative popularity Moderator: exclusivity (number of shows)

- Prior work on traditional cobranding emphasizes the benefits of exclusivity (fewer collaborating brands). Does this also hold for crossover cobranding?
- Less popular partners are often found to benefit more from cobranding. Do crossover events exhibit the same asymmetry?

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Data

6 Arrowverse series, 2012 to 2020

550 episode-level Nielsen observations

6 crossover events

2 to 5 participating shows per event

Season Year	Crossover Title	Ep. 1	Ep. 2	Ep. 3	Ep. 4	Ep. 5
2014-15	Flash vs. Arrow	The Flash S1E8	Arrow S3E8	-	-	-
2015-16	Heroes Join Forces	The Flash S2E8	Arrow S4E8	-	-	-
2016-17	Invasion!	The Flash S3E8	Arrow S5E8	LoT* S2E7	-	-
2017-18	Crisis on Earth-X	Supergirl S3E8	Arrow S6E8	The Flash S4E8	LoT S3E8	-
2018-19	Elseworlds	The Flash S5E9	Arrow S7E9	Supergirl S4E9	-	-
2020-21	Crisis on Infinite Earths	Supergirl S5E9	Batwoman S1E9	The Flash S6E9	Arrow S8E8	LoT S5E0

Series: Arrow, The Flash, Supergirl, Legends of Tomorrow (LoT), Black Lightning, Batwoman.

Model specification

$$y_{i,s,e} = \alpha_{i,s} + \beta \cdot \log(e) + \rho \cdot y_{i,s,e-1} + \mathbf{X}^{\text{pre}}{}_{i,s,e}\,\mathbf{\Gamma}_{\text{pre}} + \mathbf{X}^{\text{within}}{}_{i,s,e}\,\mathbf{\Gamma}_{\text{within}} + \mathbf{X}^{\text{post}}{}_{i,s,e}\,\mathbf{\Gamma}_{\text{post}} \cdot y_{i,s,e-1} + \varepsilon_{i,s,e}$$

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- DV: episode-level Nielsen viewership
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Main findings: pre-crossover

Pre-crossover effect

0.142

Upcoming crossover events increase viewership of the episodes before them.

× Relative popularity

−0.060

Less popular shows gain more from anticipation.

× Number of shows

−0.030

Smaller, more exclusive crossovers build stronger anticipation.

Parameter	Variable	Description	Coef.	SE	P>|z|	95% CI	
Γ_{pre}	$EpisodeBefore_{i,s,e}$	*Pre-crossover effect*: Impact of an upcoming crossover event on viewership of the current episode	**0.142**	(0.016)	0.000	0.111	0.173
	× $RelativeViewership_{i,s,e}$	Relative popularity (Pre): Influence of a show's relative popularity on the pre-crossover effect	**−0.060**	(0.002)	0.000	−0.064	−0.056
	× $NumCrossShow_{i,s,e}$	Crossover exclusivity (Pre): Influence of the number of participating shows on the pre-crossover effect	**−0.030**	(0.009)	0.001	−0.047	−0.012

Main findings: within-crossover

Immediate spike, first crossover episode

0.659

Larger for more popular shows (1.013) and for larger-scale crossovers (0.137). The scale result runs opposite to the exclusivity principle.

Carryover across crossover episodes

0.361

Weaker for more popular shows (−0.363). Crossover scale does not moderate carryover (0.013, not significant).

Reading

Popular shows draw the opening audience and depend less on momentum between episodes. Larger events open bigger.

Parameter	Variable	Description	Coef.	SE	P>|z|		95% CI
Γ_{within}	$\text{FirstCrossover}_{i,s,e}$	*Immediate crossover viewership spike*: Viewership surge in the first crossover episode	**0.659**	(0.095)	0.000	0.473	0.846
	× $\text{RelativeViewership}_{i,s,e}$	Relative popularity (Within): Influence of a show's relative popularity on the immediate crossover effect	**1.013**	(0.125)	0.000	0.769	1.258
	× $\text{NumCrossShow}_{i,s,e}$	Crossover exclusivity (Within): Influence of the number of participating shows on the *immediate crossover effect*	**0.137**	(0.027)	0.000	0.085	0.189
	$\text{LagCrossViewership}_{i,s,e}$	*Carry-over effect within crossover*: Influence of viewership in earlier crossover episodes on current crossover episode viewership	**0.361**	(0.026)	0.000	0.311	0.412
	× $\text{RelativeViewership}_{i,s,e}$	Relative popularity (Within): Influence of a show's relative popularity on the within-crossover carry-over effect	**−0.363**	(0.079)	0.000	−0.518	−0.209
	× $\text{NumCrossShow}_{i,s,e}$	Crossover exclusivity (Within): Influence of the number of participating shows on the within-crossover carry-over effect	**0.013**	(0.029)	0.661	−0.043	0.069

Main findings: post-crossover

Post-crossover retention

0.069

Episode-to-episode carryover is stronger after a crossover than before.

× Relative popularity

0.102

More popular shows keep the audience, the reverse of the pre-event pattern.

× Number of shows

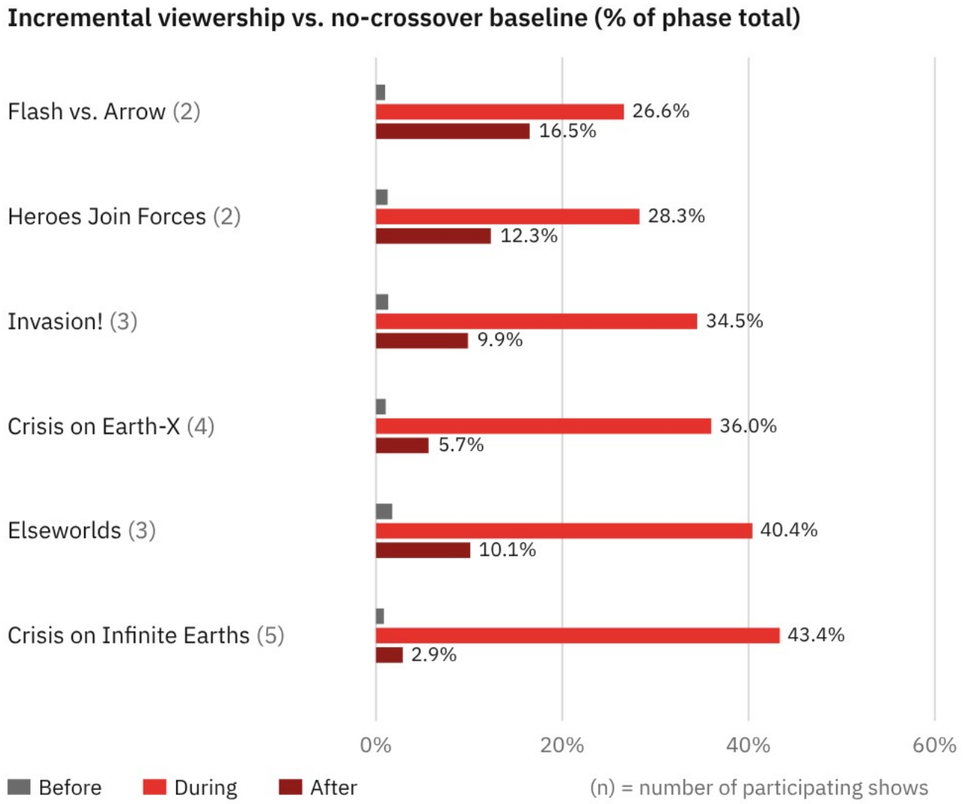
−0.024

Larger events weaken long-run retention; smaller events sustain engagement.

Parameter	Variable	Description	Coef.	SE	P>|z|		95% CI
Γ_{post}	$EpisodeAfter_{i,s,e} \cdot y_{i,s,e-1}$	*Post-crossover retention effect*: Influence of preceding crossover episode on post-crossover carryover effect	**0.069**	(0.018)	0.000	0.033	0.105
	× $RelativeViewership_{i,s,e}$	Relative popularity (Post): Influence of a show's relative popularity on the post-crossover retention effect	**0.102**	(0.028)	0.000	0.048	0.156
	× $NumCrossShow_{i,s,e}$	Crossover exclusivity (Post): Influence of the number of participating shows on the post-crossover retention effect	**−0.024**	(0.008)	0.005	−0.040	−0.007

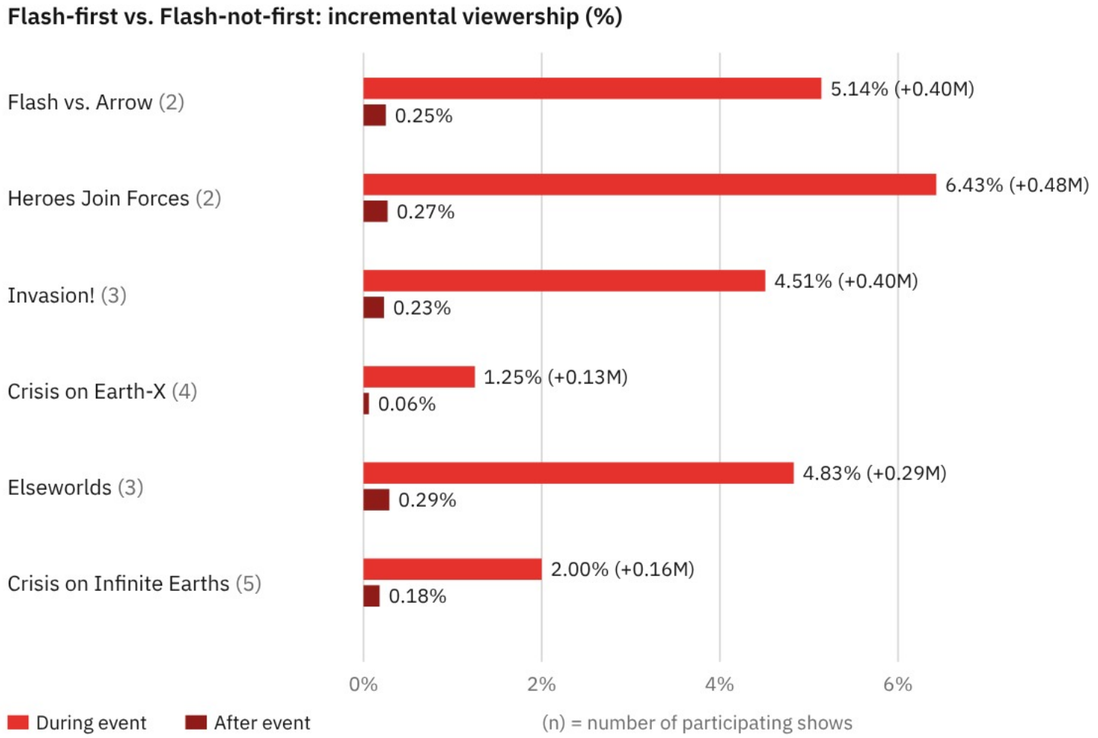
Counterfactual simulation 1: whether to run a crossover

- Crossover-related coefficients set to zero to construct the no-crossover baseline; compare with the observed trajectory.
- All six events yield positive net gains: combined **+6.0% to +12.5%** of season viewership.
- During-event gains grow with event scale (27% to 43%).
- After-event gains shrink with scale (16.5% down to 2.9%): larger events spike more but retain less.
- Largest absolute gain: *Flash vs. Arrow*, +19.6M viewers; the five-show *Crisis on Infinite Earths*, +4.9M.



Counterfactual simulation 2: which show goes first

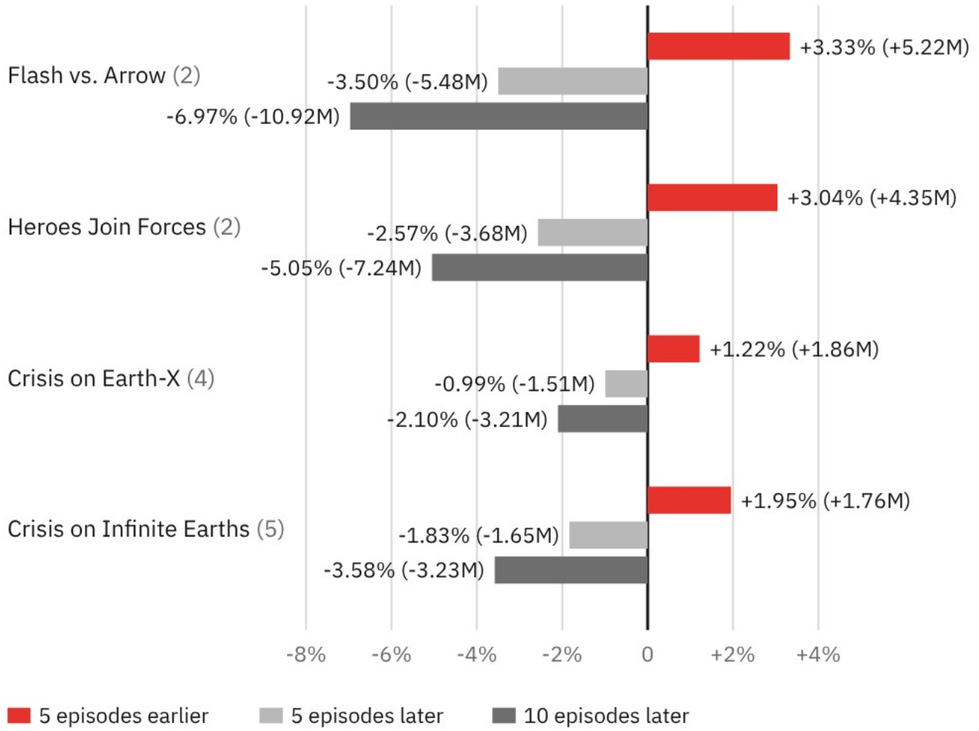
- Pairwise comparison: *The Flash* (most popular) in the first slot vs. a later slot, participating shows held fixed.
- Flash-first raises within-event viewership in all six events, by **up to +6.4%** (*Heroes Join Forces*).
- Post-event gains are small but consistently positive (up to +0.29%).
- Combined gain up to +0.51% of season viewership.



Counterfactual simulation 3: when to schedule

- Crossover moved 5 episodes earlier, or 5 and 10 episodes later, within the season.
- Earlier placement raises total season viewership by **up to +3.3%**; a 10-episode delay costs **up to −7.0%**.
- Gains are largest for two-show events; larger events move in the same direction with smaller magnitude.
- Earlier placement gives anticipation and post-event retention more episodes to act on.

Change in total season viewership vs. original schedule (%)



Invasion! and Elseworlds excluded: crossover episode numbers differ across participating shows.

Theoretical contributions

Crossover collaborations generate dynamic audience responses that unfold before, during, and after the event.

Unlike traditional cobranding, the outcome is a trajectory, not a single product-level measure.

Traditional cobranding principles do not uniformly generalize to crossover media contexts.

Exclusivity and popularity exert phase-specific, and sometimes opposing, effects across pre-, within-, and post-crossover periods.

The findings extend cobranding theory to serialized, multi-entity media collaborations.

Managerial contributions

Before the event

- Smaller and more exclusive crossovers generate stronger pre-event audience build-up.
- Crossovers can strategically boost anticipation for less popular shows.

During the event

- Launching the crossover on the most popular show increases overall audience engagement.
- Larger crossover events create stronger immediate audience spikes.

After the event

- Smaller and more exclusive crossovers produce stronger long-run post-event retention.
- Post-event retention is stronger for more popular shows.

Scale, sequencing, and timing are scheduling decisions. The estimated model can be used to evaluate a crossover plan before committing to it.

Thank you

The Viewership Effects of Crossover Cobranding

Joon Ro, Eunjoo Han, **Seoungwoo Lee**

Seoungwoo Lee
Yonsei University School of Business